

Chapter 8

Reassessing the emergence of the PIE *-*eh*₂-factitive present type

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The PIE *-*eh*₂-factitive present type is currently known under the canonical instance of **néu-eh*₂- ‘to make new’ (Hitt. *nēwahḫ-*, Lat. *(re)nouāre*, etc.), as based on the thematic adjective **néu-o-* ‘new’. The type has reflexes in Latin, Greek, Germanic, Balto-Slavic, Celtic, and possibly Tocharian, mostly in the thematic form *-*eh*₂-*je/o-* (like denominatives from abstract *-*eh*₂-stems). The Anatolian facts have confirmed that this formation was originally athematic and inflected according to the **h*₂*e*-conjugation, reflected by the *hi*-conjugation of the Hitt. *nēwahḫ-*-type, while the other Anatolian languages (Luwian, Lycian, Lydian) adopted the *mi*-conjugation of the comparable types. According to the revision of the data by Sas-seville (2021), the original function of this type in Anatolian was the derivation of denominal verbs from substantives, which was then extended to adjectives of various sorts. Several instances conclusively point to -*ah*₂-abstracts as the basis, which raises the issue of the so-called “conversion” of a noun into an athematic present as the original PIE derivational process. As a preliminary investigation, the paper reviews other alleged reflexes of this process in other present types, which seem to point to denominative formations without apparent suffix, alternatively with zero suffix, for instance PIE **g*^h*ih*₃*u*é/ó- ‘to live’, vis-à-vis the PIE adjective **g*^h*ih*₃*u*ó- ‘living’. It will be shown that these apparent cases of conversion of a noun into a verb are illusory and result from analogical developments, which presuppose the PIE existence of the “classical” full grade thematic present type, to wit **b*^h*ér-e/o-*.



1 Vedic *rāj-* ‘to rule, be king’

1.1 Verb and corresponding nouns with a lengthened grade

The Vedic verb *rāj-* (RV +) is currently glossed by an array of notions which leaves as somewhat uncertain the basic meaning, either ‘to rule, reign’, or ‘to shine, beam’, compare ‘herrschen, König sein, strahlen, glänzen’ (EWAia II 444). This semantic ambiguity is actually linked to derivational and etymological issues. The corresponding root noun *rāj-* clearly means ‘king, ruler’. It is rare as independent noun (RV 3x), while being mostly found as second compound member, in *sam-rāj-*, masc. (RV 38x, fem. *sam-rājñī-* 4x), *vi-rāj-* (7x), *sva-rāj-* (20x), *jyeṣṭha-rāj-* (2x), *eka-rāj-* (1x), *apna-rāj-* (1x), *vane-rāj-* (1x), likewise in Old Iranian, see Av. *barāzi-rāz-*.¹ Those compounds belong to the type of verbal governing compound with root nouns as second member. The normal word for ‘king, ruler’, is *rājan-*, masc. (RV 256x), nom.sg. *rājā*, acc. *rājānam*, gen. *rājñas*, dat. *rājñe*, nom.pl. *rājānas*, etc. If one adopts the current reconstruction of the root, this item can be traced back to a noun **h₃rēg-on-*, with the suffix suggesting AK inflection.² At face value, it looks like an individualizing derivative (**-on-/-n-*) besides **h₃rēg-*, which would be an abstract meaning ‘ruling, rulership’. This root noun is effectively attested as animate, and masculine, nom.sg. **h₃rēg-s*, which is reflected by RV *rāt*, matching Lat. *rēx*, OIr. *rí*, Gaul. *-rix*, but as animate, it could also be feminine, without specific marking of the gender, as testified by RV 5.46.8b. According to Weiss (2017), PIE **h₃rēg-* may have originally been a neuter root noun at the time when PIE had no feminine, meaning ‘ruling, directing’. The basis of the nasal stem was the locative sg. of this root noun: **h₃rēg-én* ‘in the power’ (> Ved. *rājān-i*, RV 10.49.4c), from which the heteroclitc neuter stem found in Avestan, *rāzara*, gen.sg. *rāzang*, instr.sg. *rašnā*, gen.pl. *rāšnqm* was backformed. The delocative animate yielded the nasal stem **h₃rēg-on-* masc. ‘being in power’ > ‘ruler’, hence Ved. *rājan-*. The *-n*-stem abstract ‘rulership’ is also the basis of **h₃rēg-n-ih₂* fem. possessive ‘having rulership’ (female) > ‘queen’, cf. Ved. *rājñī-*, OIr. *rigain*, Lat. *rēgīna* < **rēgnīnā* (as per Pinault 2014: 293–295).³

This reconstruction leaves intact the fact of the apparent coexistence of the root noun Ved. *rāj-* and of an athematic root present of the verb *rāj-*, as reflected by Ved. (RV) 3sg.act. *rāṣṭi*, which is unexpected from a synchronic point of view.

¹Survey of the RVic material by Scarlata (1999: 446–451).

²Henceforth, I use the following abbreviations for the standardly assumed PIE inflectional types: AS = acrostatic, PK = proterokinetic, HK = hysterokinetic, AK = amphikinetic, see Widmer (2004), Fortson (2010).

³Adopted with some minor differences by Weiss (2017: 796–797, 799).

Projected back into PIE, the verb **h₃réĝ-ti* has been interpreted as a denominative verb formed without suffix from the noun **h₃réĝ-* because of the meaning of the former: ‘to rule’ = ‘to be king’ (s. KEWA III: 56 and EWAia II 444 with refs.). This is of course linked to the etymology of ‘king’ itself, which connects this noun to the PIE root **h₃reĝ-* (LIV² 304–305) ‘gerade richten, ausstrecken’.⁴

In Indo-Iranian, only Ancient Vedic has the root athematic present *rāṣti*. The root thematic present is much better attested: RV *rājati*, matched by Av. *°rāzaiti*, cf. *vi-rāzaiti* ‘rules’ Yt.14.47 < Indo-Iranian **rāja-ti*. In Iranian the long vowel of the root has been extended to the verbal adjective, Av. *rāšta-* ‘directed, arranged’, OP *rāšta-* ‘straight’, MP *rāšt* ‘true’, etc.⁵ In the RV, the root thematic present *rājati* is the prevalent verbal stem (49x), subjunctive (2x), participle active *rājant-* (8x), beside inf. *rājāse* (2x). There are only a few forms of other stems, aor.act.3pl. *arājiṣuḥ* (1x),⁶ intensive pres.ptcp.act. *rārajat* (1x). Projected back into PIE, this would point to a present **h₃réĝ-e-ti*, besides **h₃réĝ-ti*, the normal root thematic present, reflected by Gk. ῥέγω, Lat. *regō*, -ere, OIr. *rigid*, Go. *rikan*. The whole issue has been treated by Strunk (1987), with further literature. Strunk’s own solution (1987: 389–390) is as follows: The root **h₃reĝ-* had an acrostatic (Narten) root present with a 3sg.act. indicative **h₃réĝ-ti*, subjunctive **h₃réĝ-e-t(i)*. The thematic root present found in most languages reflects this subjunctive. Indo-Iranian **rāja-ti* is explainable by generalization of the length of the root vowel to the subjunctive, and then to the injunctive and indicative. However, the RV subjunctive 1sg.act. *vi-rājāni* (10.159.6d = 10.174.5d, in relatively late hymns) constitutes poor evidence for a subjunctive of the athematic root present because it is most probably based on the thematic stem *rāja-* which is widespread already in the RV. As far as the verb is concerned, this doctrine has been endorsed by LIV² 305. As a minor and feeble alternative, the long grade of the verb would have its source in the nominal basis. Apparently, Strunk (1987) took the long vowel grade of the Narten present and of the root noun ‘king’ as a coincidence, even though they are from the same root. Schindler’s oral communication, mentioned in a footnote (1987: 390, fn. 3), would point, with question mark, to a root noun with acrostatic root ablaut, **réĝ-/rēĝ-* (according to the notation used by

⁴See Adams in Mallory & Adams 1997: 329–331, with previous literature. I leave aside the speculative and unsuccessful attempts to derive **h₃réĝ-* ‘king’ from a different root such as **reh₁ĝ-* or **h_xreh₁ĝ-* (more specifically **h₂reh₁ĝ-* through the connection with Gk. ἀρήγειν ‘to help’, ἀρήγων ‘helper’) by several scholars in the past; see the literature in KEWA III 50 and EWAia II 445–446. In the Indian tradition itself, Skt. *rājan-* has also been connected to roots other than *rāj-*, see Hara 1969.

⁵See further Cheung 2007: 196–198, under **Hraz-* ‘to draw a line, to direct’.

⁶See Narten 1964: 221–222, under *rāj-* ‘glänzen’.

Strunk), with further generalization of the lengthened grade, while the expected alternation is found in the heteroclitic neuter noun, OAv. *rāzarā* ‘statute, order’, besides instr.sg. *rašnā* (already quoted above) to wit **réḡ-ṛ/*réḡ-n-*, comparing (implicitly) the word for ‘liver’. This idea was later adopted in the framework of “Narten systems” favored by Schindler in the late period of his research and teaching in Vienna. Schindler assumed that some PIE roots displayed “Narten” behavior (i.e., lengthened grade of the root in place of normal full grade **e* in the strong stem, besides full grade in place of expected zero grade), both in their verbal and in their nominal forms. This is reflected, for instance, in the treatment of the derivatives of the root **(h₃)reḡ-* ‘lenken, herrschen’ by Widmer (1997: 29–31). This included the reconstruction of the root noun as an original abstract noun and of the heteroclitic neuter as **(h₃)réḡ-ṛ/*(h₃)réḡ-n-*. In this discussion, the formation of the root present Ved. *rāṣṭi* was not addressed, however, possibly because it was considered as marginal. The issue is not substantially modified by the reconstruction of the root as **h₃reḡ-*, which imposed itself in the meantime.

1.2 Explanation of Ved. *rāṣṭi* ‘reigns’

The coexistence of Ved. *rāṣṭi*, *rājati* ‘rules, reigns’ and of the noun (°)*rāj-* ‘ruler’ ‘ruler’ is puzzling because these verbs cannot be considered denominative presents in the normal sense. One would not deny that the prevalent use of RV *rāṣṭi* and *rājati* ‘to rule’, i.e., ‘to be king’ has been influenced by the meaning of (°)*rāj-* ‘ruler’, independent of whatever the original meaning of the root *rāj-* was. An alternative strategy, which was favored in the early stages of Sanskrit lexicography, is to set up two roots on the basis of the translations of the occurrences in the most ancient text, the collection of hymns of the RV. There is however no need, *pace* Grassmann (1996: 1156), to set up two homophonous verbal roots: 1. glänzen, strahlen (‘shine, be bright’) 2. herrschen, gebieten über (‘rule (over)’, governing the genitive). Both verbs would share the same present formation *rājati*, ptp. *rājant-*, so that this solution seems uneconomic. In most cases, the meaning ‘to shine, be bright’ is only inferred from the context, because the agent is a shining god (Agni, fire) or object (sun, liquid). The separation of these two meanings is actually superfluous, as can be shown by checking the majority of standard translations of the RV, which consistently situate the understanding of *rāj-* under the notions of ruling and kingship of the agent of this verb.⁷ In the most recent translation of the RV by Jamison & Brereton (2014), one finds ‘rule (over)’, ‘reign’ in most occurrences. Alternative possibilities are considered in

⁷See Geldner 1951 and Renou, EVP.

a minority of instances, and mostly between brackets: ‘rule [/shine]’ in 3.56.8b, 8.7.1c, 8.19.31d, 9.75.3d, 10.170.1d, ‘reign [shine forth]’ in 5.55.2b, ‘shine [/rule]’ in 8.60.15d, ‘shine forth’ in 8.14.10c (for the drinks of *soma*, subject of the aorist *arājiṣuḥ*, hapax, with preverb *vī*), 9.5.1b, 9.5.3b, 9.61.18b, 10.189.3a, ‘radiates’ in 9.71.7d. This survey shows that one cannot absolutely exclude the interpretation of several occurrences of *rāj-* as connected to the notion of shining (up) or beaming. This should not be taken to entail that ‘king’ itself (*rāj-*, *rājan-*) originally meant ‘the bright one’, following a by now obsolete interpretation. The meaning ‘king, ruler’ is anchored in the inherited lexicon of Indo-Aryan and is supported without any doubt by the external comparanda. However, the semantic link between PIE **h₃rég-* and the root **h₃reg-*, for which there have been somewhat different approaches, poses yet another issue which would lead us beyond the limits of the present paper. For the purposes of my argument, it is sufficient to recall that this etymological connection is accepted by a wide majority of scholars.⁸ The semantic link from ‘stretch, direct’ to ‘rule’ is certainly inherited. This was probably based on the concept of the ruler whose power and control covers a vast domain, his realm, and extends in several directions. In Ancient Vedic, the transfer ‘ruling in every direction’ > ‘shining forth’ is due to the metaphorical association of the glory and supreme power of the king with the brightness of the sun and the radiance of the weapons of the warriors. In this area of expressions of symbolic notions, one may also note the favored use of the preverbs *vī* ‘apart, asunder, in different directions’ and *sám* ‘together’ with the verb *rāj-* and with its nominal derivatives. For the occurrences in the RV where the reference to ‘shining’ seems to be more convenient and more exact than ‘ruling over’, one may also invoke some rhyming effect with the verb *bhrāj-* ‘to shine, beam, glitter’⁹ in some cases, which also forms a thematic root present, RV middle *bhrájate*, which is par excellence the verb used to describe the shining of the fire god Agni. In a somewhat conservative way, Gotō (1987: 267–271) sets up two verbs (i.e., two thematic root presents) *rājati* (1. strahlen, 2. herrschen), basically following Grassmann, but admits that the two meanings are difficult to discriminate in most instances. Moreover, there is no link between these two meanings and different verbal formations nor etymologies. The whole picture is murky.¹⁰

Returning now to the problem of derivation, the issue of *rāṣṭi* and *rājati* as anomalous denominatives is actually a pseudo-problem. The basis for the athe-

⁸As an aside, it should be recalled that the Anatolian data show that this etymon is limited to the languages of Core IE, so that **h₃rég-* cannot be deemed any longer as the unique and prototypical noun for ‘king’ in PIE.

⁹EWAia II 279–280; PIE root **b^hreh₁g-* in LIV² 92.

¹⁰See also the discussion by Scarlata (1999: 445–446).

matic root present is small indeed, since it is only found in two occurrences in the 3sg. active, once in the indicative, and once in the injunctive:

- (1) RV 1.104.4b
rāṣṭi *śúraḥ*
 reign.3SG.PRS.IND.ACT champion.NOM.SG.M
 ‘The champion reigns.’
- (2) RV 6.12.5cd
sadyó *yáh* *syandró* *viṣito*
 immediately who.NOM.SG.M hasty.NOM.SG.M loosened.NOM.SG.M
dhāvīyān | *ṛṇó* *ná tāyúr* *āti*
 hurrying.COMP.NOM.SG.M debtor.NOM.SG.M like thief.NOM.SG.M across
dhán(u)vā *rāt* ||
 wasteland.ACC.PL.N reign.3SG.PRS.INJ.ACT
 ‘He [Agni] who immediately, when unloosened, streams ever faster, like a debtor (turned) thief he has headed straight across the waste places.’
 (transl. Jamison & Brereton 2014: 786).

By chance (or by a single poet’s choice?), the same hymn features one of the few (three in total) occurrences of the nom.sg. *rāt*, 6.12.1a, at the same place in the pāda. Besides *rāt*, the simplex noun *rāj-* has no other forms in the RV. Therefore, it seems better to explain this root athematic present as an innovation within Ancient Vedic, which remained unsuccessful compared to *rājati*, inherited from Indo-Iranian **rāja-ti*. One should find a way to explain both presents, without resorting to the notion of denominative formations. My scenario is as follows. The Indo-Iranian thematic present was inherited parallel to the formations attested in other IE languages and originally with normal full grade of the root: **rāja-ti* < **h₃rég-e-ti*, ptcp. **rājant-* < **h₃rég-o-nt-*. Those forms were remade as **rāja-ti*, ptcp. **rājant-* under the influence of the nasal stem **rājan-* < **h₃rég-on-*, and of the second compound member *°rāj-* < **°h₃rég-*. The two latter items had been independently inherited. The nasal stem was derived via a delocalative derivation as outlined above and became firmly entrenched as the designation of the ruler at some stage. The process of reshaping the verbal stem hinges on two crucial points. The first point is the equivalence of the second member of a verbal governing compound with a participle and an agent noun, to wit ‘ruling over’. The second point is the crucial role of the present participle as indicative of the structure of the corresponding present stem. This will also be relevant in the interpretation of the thematic present **g^hih₃u^hé-ti*, which we will discuss in the second section of this paper.

The king's function was correlated with the territory where his power was in force and where the clans (Ved. *vis-*) under his authority reside, live and prosper in peace, a notion covered in Ancient Vedic by the root *kṣay-/kṣi-* 'to dwell' and its derivatives, see especially *kṣéma-* masc. 'peace, peaceful dwelling'. The Vedic term *rāṣṭrá-* nt. (RV 10x) 'realm, kingdom' (basis of *rāṣṭrī-* 'ruler' 3x) has been formed after the model of *kṣétra-* nt. 'field, ground, country' (RV 20x, plus in several compounds), the latter being inherited, cf. OAv. *šōiθra-* 'district, country'. The phrase RV *kṣétrasya páti-* is matched by Av. *šōiθrahe paiti-* 'lord, master of the field' = 'king'. This lexical configuration yielded the causal force for the association between noun and verb. The verb *kṣay-/kṣi-* has a root athematic present inherited from Indo-Iranian and from PIE, namely RV 3sg.act. *kṣéti*, 3pl. *kṣiyánti*, etc. (LIV² 643, root **tkei-*). Based on these inherited lexemes, a proportion took shape: noun *kṣétra-* : root present *kṣéti* :: *rāṣṭrá-* : X = *rāṣṭi*.

A proportional analogy such as this suffices to account for the root athematic present and de facto eliminates one of the favorite instances of an alleged denominative verb without suffix, i.e., of conversion of a noun into a verb. One further advantage of the whole solution is to dispense with the otherwise interesting idea of a "Narten" root in the case of the root **h₃reǵ-*. I may recall from personal recollection that Schindler was forcefully opposed to the idea of denominative verbs without marking of derivation by a suffix. The preceding paragraphs should be understood as a first modest answer to this recurring problem.

2 The PIE thematic present **g^uih₃ué-ti*

2.1 A PIE **-u*-present

A PIE thematic present can be safely reconstructed on the basis of several IE languages: Ved. *jívati*, YAv. *juuaiti*, Lat. *uīuit*, OPr. *giwa*, Latv. *dzīvu*, OCS *živŏ*. As for the root grade, Toch. B *śāw-*, stem CToch. **śāw'ā-* > Toch. B **śāyā-*, pres. act. 3sg. *śaim* (besides CToch. **śāwæ-* in Toch. A act. 3pl. *śāweñc*) is ambiguous, since **śāw-* may reflect **g^uioh₃u-* as well as **g^uih₃u-*.¹¹ Gk. ζῶω has probably been reshaped with full grade on the basis of the root athematic aorist 3sg.act. **g^u(i)éh₃-t* (cf. Gk. ἐβίων). In Vedic, the retraction of the accent to the root was due to the restructuring of the neo-root *jīv-* as a plain thematic present.

The mainstream account (e.g., LIV² 215–216, Jasanoff 2003: 142) assumes a PIE thematic present which originated as a thematization of an athematic **-u*-present **g^uieh₃-u-/g^uih₃-u-*. This type followed the **h₂e*-conjugation and hence had a

¹¹See Malzahn 2010: 916–919 with previous literature.

3sg. $*g^u\check{i}eh_3-u-e$ that was the nucleus of the later thematic paradigm (Jasanoff 2003, 2023). To this scenario one can object that there are no reflexes of an athematic $*-u$ -present from the root $*g^u\check{i}eh_3-$, contrary to the reflexes of other verbs: $*terh_2-u-$ ‘overcome, cross over’ (Hitt. *tarḫu-*, 3sg.act. *taruḫzi*, *tarḫuzzi*, Ved. mid. *tarute*, thematic Ved. *túrvaṭi*, YAv. *tauruua-*), $*leh_2-u-$ ‘pour’ (or $*leh_3-u-$, as per Melchert 2011), Hitt. *laḫu-* (3sg.mid. OH *laḫuwāri*, ptcp. *laḫu(w)ānt-*), $*uer-u-$ ‘ward off’, $*uel-u-$ ‘turn’, etc.¹²

2.2 Questioning the theory of conversion

An alternative theory that was entertained for a long time is that $*g^u\check{i}h_3u\acute{e}-ti$ was a denominative present based on the adjective $*g^u\check{i}h_3u\acute{o}-s$ (nom.sg.masc.) reconstructed for Core IE as well: Ved. *jīvā-*, Av. *juua-*, Lat. *uīuus*, OIr. *biu*, Go. *qius*, Lith. *gývas*, Latv. *dzīvs*, OCS *živŭ*.¹³ The infinitive Lat. *uīuere*, Ved. *jīvāse* reflects a stem $*g^u\check{i}h_3-ue/os-$ (underlying also Toch. B *śaiṣṣe* ‘world’, A *śoṣi* ‘people’)¹⁴ of the abstract based on this adjective.

This pair of adjective and verb has been put into service by Villanueva Svensson (2021: 273–277), claiming that the thematic present issued from the thematic adjective by a process of “conversion”, which may have been a regular derivational process at an early PIE stage:

- (3) adj. $*g^u\check{i}h_3-u\acute{e}/\acute{o}-$ → verb 3sg. $*g^u\check{i}h_3-ue-t(i)$, 3pl. $*g^u\check{i}h_3-u\acute{o}-nt(i)$.

According to Villanueva Svensson, this pair is but a peculiar instance, which has been exceptionally preserved in Core IE, of the derivation by conversion of a thematic present from a thematic adjective (2021: 273–277), which was a purely nominal formation, not based on an alleged, but actually non-existent $*-u$ -present (with strong stem $*g^u\check{i}eh_3-u-$). This pair reflected a standard process which is hypothetically assumed for early Core IE, through which the simple thematic present of the type $*b^h\acute{e}r-e/o-$ ‘to carry’ was derived from a reconstructed verbal adjective $*b^her-\acute{o}-$ ‘carrying’ (Villanueva Svensson 2021: 277–284). Consequently, the whole root thematic present type emerged *after* the separation of Anatolian from PIE and became widely productive in Core IE. The latter point is the major claim of his paper, because Villanueva Svensson dismisses previous theories

¹²See Jasanoff 2003: 141–143; 2023: 63–66 for these examples and a few others. *Pace* Jasanoff 2023: 66, Toch. B *śaumo* ‘man, human’ (A *śom* ‘boy’) cannot be projected back to PIE $*g^u\check{i}eh_3u-m\acute{o}(n)$ based on the suffixed root. This item ought to be explained as created at the CToch. stage, as per Pinault 2021: 119–120.

¹³See Tichy 2004: 55, with part of the previous literature.

¹⁴As per Pinault 2008: 234, 2021: 117.

about the place of the deverbative type **b^hér-e/o-* in the PIE verbal system (2021: 264–273).

The same process is then found in other pairs consisting of a thematic adjective and a thematic root present, e.g. Gk. λευκός ‘white’ (**leuk-ó-*): Ved. *rócate* ‘shines’ (**léuk-e-to+i*), yielding the thematic root present through additional adjustments of accentuation to obtain an intermediate **léuk-o-* ‘shining’ that could serve as the basis for conversion. However, Villanueva Svensson fails to account for the full grade and the root accentuation of the **b^hér-e/o-* present type as well as the typical **-e/o-* alternation of the thematic vowel in the inflection, except through circular and *ad hoc* hypotheses (such as assuming that the original accentuation of the adjective **leuk-ó-* > Gk. λευκός was ***léuk-o-*, just for complying with the accent of the present stem **léuk-e/o-*, cf. Villanueva Svensson 2021: 281–282, and fn. 43).

An earlier variant of the conversion account for the verb ‘living’ has been formulated by Tichy (2004: 55) plainly as follows: “3.Sg. Ind. Präs. uridg. **g^uih₃-ue-ti* ‘ist **g^uih₃uó-*, lebt.” This is however a mere pedagogical paraphrase instead of an explanation, which leaves out the real nominal form that could be used as a predicate in a nominal sentence, the nom.sg. **g^uih₃uó-s*, in favor of the transposition of the thematic stem. Tichy quotes Rix (1994: 79), who pushes back the phenomenon into PIE: “hocharchaische Denominativbildung mit Nullsuffix”.

The basic problem is: What does “conversion” really mean? In other words, how would it work in syntactic and morphological terms? Does it imply that PIE was not a fully-fledged inflectional language? Under this account, there is no clearcut link through reinterpretation of a predicate thematic stem, with nominal endings, nom.sg. **^o-s* (animate) or **^o-m* (neuter) as verbal 3sg.act. **-e* (active or proto-middle; active **^e-t*, middle **^e-to* (injunctive). How could this gap be bridged? This cannot be understood remotely in the spirit of “Watkins’ Law” (Watkins 1969), according to which a form **b^hér-e* with “proto-middle” 3sg. ending was reinterpreted as a thematic stem with zero ending, triggering the generalization of the stem **b^hér-e-*.¹⁵

The **^e* form of the thematic vowel did not surface in the nominal inflection except for the vocative and possibly in some adverbial forms. Nobody has claimed that the thematic inflection started from the imperative, while taking for granted that the imperative with zero ending in the second singular active could be akin to a vocative (and vice versa). Nor is there any basis for setting up a predicative adverb ***b^her-é* ‘while carrying’, which is mentioned here only for the sake of the

¹⁵See Jasanoff 2003: 224–227. Watkins’ “law” was used as powerful tool for interpreting various verbal formations, see Joseph 1980 and Janse 2009 with further references.

argument. Therefore, it appears that the theory of conversion cannot be pushed very far, since it entails many additional speculations.

Jasanoff (2023: 74–77), on the other hand, argues that PIE did possess the full grade thematic present type, which did not arise through conversion but through a process of reanalysis of the “proto-thematic” 3sg. active ending **-e* of the **h₂e*-conjugation, coexisting with active (transitive) 1sg. **-e-h₂e*, 2sg. **-e-th₂e*, later recharacterized as injunctive 3sg.act. **-e-t*, hence indicative **-e-ti*. There are instances of standard root thematic presents already in Anatolian, not to speak of the numerous complex suffixed thematic presents, in **-īe/o-*, **-skē/o-*, **-eje/o-* with active thematic inflection. As for the above pair itself, the scenario developed by Jasanoff (2023: 62–71) would take the two items as independent formations from the **-u*-present which followed the **h₂e*-conjugation (type **térh₂-u-e*, with *-i* added to the ending in the active, as per Jasanoff 2003: 141–143): The present **g^hih₃-ue/o-* arose through thematization based on the 3sg.act. **g^hih₃-u-e(+i)* (for expected ***g^hieh₃-u-e*, presupposing levelling of the root ablaut) and was characteristically active, while the adjective was presumably a kind of thematic verbal adjective based on the regular weak stem **g^hih₃-u-*. The latter point remains somewhat in the dark, if not taken for granted.

2.3 An alternative derivation with a verbal adjective as intermediate item

In the following, I will try a different course, based on well-known facts of nominal derivation and semantics. PIE **g^hih₃-uó-* ‘living’ was a derivative from the root **g^hieh₃-* by means of the archaic verbal adjective-forming suffix **-uó-*,¹⁶ which can be compared to its semantic opposite **mṛ̥-uó-* ‘dead’ > OIr. *marb* from the root **mer-* ‘to die’, replaced by **mṛ̥-tó-* > Ved. *mṛ̥tá-*, Av. *mərəta-*, Arm. *mard*, etc., with (parallel) contamination into **mṛ̥-tuó-* > Lat. *mortuus*, OCS *mǫrtvŭ*. The development of **g^hih₃-uó-* should therefore be viewed in the context of the semantic field which covers life, old age and death. The root **g^herh₂-* ‘to wear out, age, make look older’ (LIV² 165) formed a plain thematic verbal adjective PIE **g^herh₂-ó-* ‘decayed’ (Arm. *cer*) and a parallel athematic synonymous **g^herh₂-ont-/g^herh₂-nt-* > Ved. *járant-* and *jurat-* (dat.sg. *juraté*, gen.pl. *juratám*), fem. *járatī-*, Gk. γέρων, γέροντ-, Oss. *zæronð* ‘old, decayed’, featuring AK inflection, and belonging to a Caland system (cf. the neuter abstract **g^herh₂-s* > Gk. γέρας shifted to ‘honor, perk’ and remade as γῆρας under the influence of the

¹⁶This is more cogent than resorting to the oppositional suffix found in **deks(i)-uó-* ‘right’ (Gk. δεξι(ς)ός, Go. *taihswō*), since the basis of this term was not a verbal root.

aurist ἔγρηπα, as per Stüber 2002: 84).¹⁷ The idea of **ĝérh₂-ont-* as the participle of a Narten verb, present or aorist, as per Widmer 1997: 21 (see also LIV² 165, n. 2) does not withstand scrutiny. The Ancient Vedic verbal system included the transitive-factitive present *járatī* ‘to make old’, besides intransitive stems such as the present RV *júryati*/AV *jíryati* ‘to grow old, become frail’ and the stative perfect RV participle act. *jujurvān*, *jujurús-*, AV *jajára* ‘ist gealtert’. One cannot reconstruct any PIE aorist of this verb. An **(o)n-*stem is reflected by the Toch. B stem *śrānā-* masc. ‘adult man’,¹⁸ nom.pl. *śrāy*, obl.pl. *śrānām*, gen.pl. *śrānāmts*; *śrāy* < **śārāyā* < **śārānā* < **ĝérh₂-ōn-es* (alternatively, but less likely, **ĝérh₂-n-es*), presupposing as stem **ĝérh₂-ōn-* < **ĝérh₂-o-on-* ‘the old one’, with an individualizing suffix based on the thematic adjective mentioned above.

After the model of **ĝérh₂-ont-*, reanalyzed as **ĝérh₂-o-nt-*, parallel to the synonymous **ĝérh₂-ó-*, **ĝérh₂-ōn-*, a remodelled form **g^hih₃uó-nt-* meaning ‘the living one’ was created, opposed to **mér-to-* ‘the dead one’ (Ved. *márta-* masc.) ‘mortal, human’ and **mertōn-* < **merto-on-* (Av. *marātān-*, masc. ‘human’, nom.sg. *marātā*, nom.pl. *marātānō*). At a later stage, when the verbal adjective formation in **-uó-* had been replaced by **-tō-* and **(e/o)nt-*, **g^hih₃uónt-* could be reanalyzed as **g^hih₃uó-nt-*, the participle of a thematic verbal stem, whence the present injunctive act. 3pl. **g^hih₃uo-nt*, 3sg. **g^hih₃ue-t*. The final stage of this process presupposes the PIE existence of root thematic presents of the “classical” type, to wit **b^hér-e/o-*, injunctive act. 3sg. **b^hér-e-t*, 3pl. **b^hér-o-nt*, participle **b^hér-o-nt-*. If not for this precise root, one may surmise that this type existed in PIE, see, e.g., the **-je/o-* and **-ske/o-*presents with “complex” thematic suffixes. The PIE existence of full grade thematic presents of the type 3sg.act. **b^hér-e-ti* has recently been supported by new independent arguments,¹⁹ even though the presents of this type were not as abundant at the early stage as they were later in Core IE.

To conclude, the thematic present **g^hih₃ué/ó-* was wholly analogical, while being based ultimately on an adjectival stem. This derivation offers an additional instance of creating the inflectional forms of a finite present stem based on a verbal adjective reinterpreted as participle. My scenario thus differs markedly from the recent accounts of the pair **g^hih₃-uó-* ‘alive’ : **g^hih₃-ue/o-* ‘to live’ by Villanueva Svensson (2021) and Jasanoff (2023).

¹⁷Consequently, the routine reconstruction of an acrostatic neuter **-s-*stem, as per Widmer (1997: 20, 2004: 50) and other scholars, is not necessary.

¹⁸Pinault 2008: 441, 484, differently Adams 2013: 705.

¹⁹See Jasanoff 2023: 72–78.

3 PIE **néu-ah*₂- ‘to make new’

3.1 Brief review of the Anatolian evidence

The type Hitt. *nēwahḫ-* ‘to make new’ has been much discussed in the literature.²⁰ The Anatolian evidence has recently been reassessed by Sasseville (2015 and 2021: 16–77). The type was productive in Hittite in forming deadjectival factitives from thematic and other kinds of stems: *nēwahḫ-* ‘to make new’ : *nēwa-* ‘new’, *maršaḫḫ-* : *marša-* ‘unholy’, *dannattaḫḫ-* : *dannatta-* ‘empty’, *ḥappinaḫḫ-* : *ḥappina-*, *ḥappinant-* ‘rich’, *šuppiyaḫḫ-* : *šuppi-* ‘pure’, *tepawaḫḫ-* ‘to make little’ : *tēpu-* ‘little’; from substantives, e.g. *armaḫḫ-* ‘to make pregnant’ < ‘make round like the moon’ : *arma-* c. ‘moon’, *luriyaḫḫ-* ‘to humiliate’ : *luri-* c. ‘humiliation’, *šagiyahḫ-* ‘to give a sign’ : *šagāi-* c. ‘sign’. This type belongs to the derivational set of deadjectival verbs together with fientive *-ēšš-* and factitive *-nu-* verbs.

The *ahḫ-* type followed the *hi*-conjugation in Hittite (OH and MH), which was later replaced by the *mi*-conjugation. By contrast, in the other Anatolian languages, the *mi*-conjugation prevails, with non-lenited endings in the corresponding presents in Luwian /-a-/, Lycian -a- and Lydian -a-. Those verbs are mostly derived from substantives, rarely from adjectives: HLuw. *tabariya-* ‘to command’ : *tabariya-* c. ‘command’, Lyc. A *prñnawa-* ‘to build (a grave-house)’ : *prñnawa-* c. ‘grave-house’, Lyc. B *muwa-* ‘to strengthen’ : coll. *muwa* ‘might, power’, Lyc. B *qala-* ‘to build a precinct’ : *qala-/qala-* c. ‘precinct’, Lyc. A *smma-* ‘to bind, oblige’ : *smma-* c. ‘binding, obligation’, Lyd. *pita-* ‘to make a donation’ (‘to make pay’ according to Garnier, p.c.) : **pita-* c. ‘donation’, cf. Lyc. A *pijata-* c. ‘gift’, Hitt. *piyetta-*, *pitta-*, coll. ‘allotment, payment’. They could also be derived from adjectives or agent nouns: CLuw. *tannatta-* ‘to make empty, devastate’ : *tannatta/i-* ‘empty’, CLuw. *arkammanalla-* ‘to make someone into a tributary’ : *arkammanalla/i-* ‘tribute-bearing’, Lyc. A *kumaza-* ‘to perform a sacrifice’ < ‘to act as a priest’ : *kumaza/i-* ‘sacrificing priest’, HLuw. *hantawatta-* ‘to act as a king’ : *hantawatt(i)-* c. ‘king’. Those agent nouns actually reflect formations in *°ā* < **-eh*₂, the individualizing suffix, e.g., Lyc. **-za-* < **-tiāh*₂, derived from secondary adjectives in **-tjo-* (see Sasseville 2017, 2018).²¹

²⁰Cf. Oettinger 1979: 155, 240, 247, 253, 455–458, Melchert 1997: 132, Jasanoff 2003: 139–141. These references are not intended to be exhaustive. See also the introduction to this volume for an overview.

²¹Following Melchert 2014, see further below.

3.2 Considering the issue of conversion

The usual comparanda for the type are the *-ā-factitives in Latin, Greek, Germanic, Balto-Slavic, and Celtic. This is often summarized by the equation Hitt. *nēwahh-*, Lat. *(re)nouāre*, Gk. *νέω* ‘to plough up anew’, OHG *niuwōn*, despite some secondary restructurings. Thus, Gk. *νέω* is actually associated with *νείος* fem. ‘new-ploughed land’ (< **neuijō-*), not with the adjective *νέος* ‘new’ and OHG *niuwōn* is based on the alternative Germanic adjective ‘new’, OHG *niuwi* (< **neu(i)jo-*, cf. Ved. *náv(i)ya-*, doublet of *náva-*). Nonetheless, the derivational process can be summarized by the following scheme in PIE:

- (4) thematic adjective **néu-o-* ‘new’ →
verb **néu-e-h₂-* ‘to make new’, 3sg.act. **néu-e-h₂-e(i)*, 3pl.act. **néu-e-h₂-nti*

This entails that the thematic vowel of the base took on the *-e-grade, assuming that the verbal suffix is reconstructed as plain *-h₂-. Alternatively, the suffix *-eh₂- would be substituted for the final *-o- of the base. This moot point is ordinarily not considered in the scholarship, even though it has significant consequences for ascertaining the origin of the formation.

This present formation was subsequently thematized to *-eh₂-īe/o-, which is found in Latin, Greek, Balto-Slavic, Vedic, and possibly Tocharian,²² as denominatives from reflexes of PIE *-ah₂-stems. This type became quite productive in several languages. One may leave aside henceforth the case of the alternative thematization in *-eh₂-e/o-, which became productive in Balto-Slavic.²³ This type has left reflexes in Aeolic and Mycenaean Greek contract verbs, pointing to an originally athematic 3sg.act. **neuah₂ei*, cf. Lesb. -αι as 3sg.act. in the athematic paradigm of contract verbs (with 1sg.act. -ᾱμι instead of -ᾰω), and Myc. *te-re-ja*, 3sg.act. present < **teleh₂ai*, from the verb **teleh₂ā-* ‘to accomplish, fulfill one’s obligations’ (Rau 2009). The basis can be either the adjective **teleh₂io-*, a syncope form of **teleh₂io-* > Gk. *τέλειος* ‘perfect, accomplished’ (derived itself from the neuter s-stem reflected by *τέλος* ‘service, obligation’), or an abstract **teleh₂īā* < **teles-īah₂* (Rau 2009: 181, fn. 4). The Myc. infinitive *te-re-ja-e* < **āhen* (cf. Lesb. -ᾰν) has the expected form for an athematic stem, since the ending may reflect ultimately a virtual locative sg. **ah₂-sen*, similar to other enlargements of abstracts, such as the well-known types *-ah₂-ter/n-, *-ah₂-uer/n-.²⁴

²²Pinault 2008: 579, further discussion in Malzahn 2010: 393–402.

²³See Jasanoff 2003: 140–141; 2017: 200–201; Villanueva Svensson 2020: 393–395.

²⁴Nussbaum 1986: 32–34, 135, 198, fn. 4; Rieken 1999: 347–354, 380–382, 414–417.

The verbal suffix $*-h_2-$ (after thematic stems) with factitive value has not been accounted for from the etymological point of view. There is no comparable suffix in the verbal derivational system of PIE. The segmentation itself (as $*-eh_2-$ or $*-e-h_2-$ if the basis was a thematic adjective) remains elusive. If one assumes the “conversion” of a nominal stem in $*-(e)h_2-$ as the basis of a verbal stem, it would be required to ascertain what the underlying nominal derivative was, e.g., an abstract/collective neuter in $*-eh_2-$ ($*-e-h_2-$) or an individualizing animate in $*-eh_2-$.²⁵ This issue would not matter so much for scholars (such as one anonymous reviewer of the present paper) who consider “that at an early stage of Indo-European nominal and verbal inflection differed only in the endings, but not in the stem formation”, so that the suffix $*-eh_2-$ could form both nominal and verbal stems. This hypothesis is based on a very simplistic and glottogonic view of the morphological reconstruction. In any case, it would not account for the $*h_2e$ -conjugation of the PIE $*néu-ah_2-$ present type, nor for its factitive meaning. The present stems in $*-i-$ and $*-u-$ cannot be straightforwardly identified with suffixed nominal stems either, which feature several inflectional patterns which are not all found in the verbs. In addition, this approach ignores the existence of denominative presents in PIE and in the IE daughter languages, as well as of verbal adjectives provided with specific and different morphemes. I will therefore not consider it further.

As shown by Sasseville (2017: 287–292; 2021: 69–70), the original Anatolian function of verbal $-aḥḥ-$ was the derivation of verbs from substantives, which was then extended to adjectives, as in the Hitt. $nēwahḥ-$ type. Several instances conclusively point to $*-ah_2-$ abstracts as the base, for instance Lyd. $pita-$ ‘to make a donation’ from a $*-tah_2-$ stem, Hitt. $waštaḥḥ-$ ‘to sin, to offend’ vs. $waštai-$ c. ‘sin’, CLuw. $wašta-$ nt. ‘sin’, Hitt. $armaḥḥ-$ ‘to make round like the moon, to impregnate’ vs. $arma-$ c., CLuw., HLuw. $arma-$ c. ‘moon’, Lyc. A $arṁma-$ c. ‘moon-god’ < stem in $*-mah_2-$. Concerning the latter example, I follow the derivation proposed by Sasseville (2017: 282–284; 2021: 69–71), which complies with the expected semantic evolution of a verb of the factitive Hitt. $°aḥḥ-$ type. Therefore, this account is certainly to be preferred to the one suggested by one anonymous reviewer for the original meaning of this verb: ‘to bring to the months’, or perhaps ‘to bring to the month (of birth)’. This remains arbitrary, and at variance with the factitive meaning warranted by other instances of $°aḥḥ-$ verbs which are clearly based on substantives. There is no support for the alleged directive (or allative) meaning which would be taken by the basis of this denominative formation.

²⁵Cf. Melchert 2014, Nussbaum 2014: 274–275, 306, and Fellner & Grestenberger 2016 on $*-(e)h_2-$ as marker of substantivization.

In addition to these factitive verbs which were apparently derived without any overt suffix, Anatolian had normal denominatives formed with the suffix **-je/o-*, featuring essive value: e.g. Hitt. *armā(i)-* < **ah₂-je/o-* ‘to be round like the moon’ > ‘be pregnant’, contrasting with *armahh-* ‘to make round like the moon’, quoted above.²⁶

Sasseville (2015: 292–293) assumes the “conversion” of **-ah₂-*abstracts into factitive **-ah₂-*verbs, challenging the standard view of the *nēwahh-* type as derived from thematic adjectives, for Anatolian as well as for PIE. The derivation from thematic adjectives would be a development specific to Hittite within Anatolian and independently of Core IE. The process of restructuring of a nominal suffix into a verbal suffix is taken for granted, without setting up any intermediate step. This approach is endorsed by Villanueva Svensson (2020: 391–392, 2021: 275–276) because it provides welcome support to his own account which argues for the derivation of the full grade thematic present type from thematic verbal adjectives. This entails that there was originally no proper verbal suffix in verbs with a 3sg.act. **-ah₂-e*, which followed the **h₂e*-conjugation. One may surmise that the thematic renewal as **-ah₂-e/o-* found in Balto-Slavic issued originally from the reinterpretation of this 3sg. active form. The missing link could be a further individualizing derivative in **-on-* or **(e/o)nt-*, the latter being reinterpreted as a participle. This would not explain the original **-h₂e* conjugation, however. Moreover, such secondary derivatives of abstracts or animate substantives in **-ah₂-* did not leave any reflex in unmistakable form. The case of adj. **g^hih₃-uó-* ‘living’ vs. verb **g^hih₃-ué-ti* ‘(s)he lives, is alive’ does not add support to the “conversion” account because this present *does* feature the normal thematic conjugation of the **b^hér-e-ti* type; see Section 2.3 above.

3.3 The Vedic evidence

In previous discussions of possible comparanda of the Hitt. *nēwahh-* type, Indo-Iranian has not been mentioned so far. However, some intriguing evidence in Ancient Vedic can be detected: The present RV *vr̥ṣāyāti/-te* ‘to make rain’ (3x), which must be distinguished from *vr̥ṣāyāte* (RV 7x, ptcp. *vr̥ṣāyāmāṇa-* 2x) ‘to behave like a bull, act the bull’, denominative from *vr̥ṣan-* masc. (see also the parallel adjective *vr̥ṣayú-* ‘acting the bull’ in the herd among the cows, RV 9.77.5d). The structure of the former item has not yet been cogently accounted for. The reading of the occurrences does not leave any doubt:

²⁶As argued cogently by Sasseville (2015: 287–288).

- (5) RV 10.98.1d [to Brhaspati]

sá *parjānyam śamtanave vṛṣāya* ||
DEM.PRON.NOM.SG.M P.ACC.SG S.DAT.SG rain.make.2SG.IMP.ACT

‘Make Parjanya rain for Śamtanu.’ (in a plea for rain by Devāpi, priest and poet. Transl. Jamison & Brereton 2014: 1555)

- (6) RV 9.71.3ab

ádribhiḥ sutáḥ pavate gábhast(i)yor,
stone.INS.PL pressed.NOM.SG.M purify.3SG.PRS.IND.MID hand.LOC.DU
vṛṣāyáte nábhasā vépate
rain.make.3SG.PRS.IND.MID cloud.INS.SG tremble.3SG.PRS.IND.MID
matí |
thought.INS.SG

‘Pressed by the stones, he (Soma) purifies himself between the two hands. With his cloud he makes rain (the rain = his juice); he trembles (in poetic inspiration) with his thought.’ (differently from Jamison & Brereton 2014: 1304, who translate ‘he acts the bull [/rains]’).²⁷

- (7) RV 10.44.4ab

evá pátiṃ droṇa-sácam śáquetasam,
thus lord.ACC.SG cup-companion.ACC.SG like.minded.ACC.SG
ūrjá skambháṃ dharúṇa á
nourishment.GEN.SG prop.ACC.SG support.LOC.SG PRVB
vṛṣāyase |
rain.make.2SG.PRS.IND.MID

‘Just in this way you [Indra] drench yourself in the lord [= soma], the companion of the cup, the like-minded one, the prop of nourishment upon its support.’ (transl. Jamison & Brereton 2014: 1449).

This present stem provided the basis for the secondary *-iṣ-*aorist *ávṛṣāyiṣata* (VS II.31), following the pres. imperative *ávṛṣāyadhvam* (cf. Narten 1964: 62, 292), both with preverb *á*, as in RV 10.44.4b. The formation of the present is not discussed by Narten.

This present *vṛṣāyáti/-te* can hardly represent a formation of the *grbhāyáti* type, based originally on nasal infixed present, to wit *grbhñáti* from *grabhⁱ*- ‘to

²⁷The first option corresponds to Geldner’s choice: “durch die Regenwolke wird er wie ein Bulle, durch die Dichtung wird er beredt” (Geldner 1951: III, 63). See further discussion by Renou (EVP VIII 83 and IX 81), who notes that the reference to the rain is implied by *nábhas-* as complement, evoking the cloud.

seize', with $\bar{a}y\acute{a}$ < *-aH-y\acute{a}- < *-ḡH- $\dot{i}é/o$ -,²⁸ since *varṣ-/vrṣ-* (*aniṣ* root) did not form a -*nā*-present. Nor can it be a plain essive or fientive denominative from a noun **vr̥ṣā-* (or **vr̥ṣa-*) meaning 'rain', because it has factitive meaning.²⁹ This verb did not survive long, since it was in competition with the regular causative *varṣáyati* 'to make rain' (RV 3x, participle *varṣáyant-* 1x, with complement *dyām* 'sky' or *vr̥ṣtīm* 'rain'), besides the intransitive pres. *várṣati* 'to rain'.³⁰

Instead, I propose that *vr̥ṣā-yá-ti* replaced a factitive athematic present **ur̥ṣā-* < **h₂ur̥seh₂-*. The latter cannot be based on an adjective **h₂ur̥s-ó-* 'rainy, raining', which is unattested. The basic nominal derivatives of the PIE root **h₂uers-* are well known (see Nussbaum 2017: 255–256): agentives which have been variously substantivized, namely **h₂uórs-o-* or **h₂uors-ó-* (Hitt. *warša-* c.), **h₂uors-ó-* or **h₂uers-ó-* (Ved. *varṣá-* nt.), collective **h₂uersé-h₂-* (Ved. *varṣá* 'the rains', 'rainy season'), abstract **h₂uérse-h₂-* (Gk. $\acute{\epsilon}\acute{\epsilon}\rho\sigma\eta$, $\acute{\alpha}\acute{\epsilon}\rho\sigma\eta$ 'dew'). The potential stem **h₂ur̥s-éh₂-* could be a primary derivative of the type Gk. $\phi\upsilon\gamma\acute{\eta}$, Lat. *fuga* 'flight' or a replacement of the expected root noun **h₂uér-s-/h₂ur̥s-* 'raining' reflected by RV *prāvr̥ṣ-* fem. (2x) 'beginning of the rainy season', derived adj. *prāvr̥ṣīṇa-* (1x).³¹

In order to explain the factitive present from an abstract, the best way lies in the syntactic reconstruction of an intermediary nominal sentence:

- (8) dative sg. in predicative function **h₂ur̥séh₂-ei* '(s)he/it (is) for the raining'
> '(s)he/it makes rain'.

Next, the sequence could be resegmented as a 3sg. present **h₂ur̥séh₂-e+i* with a primary ending, hence as a verbal stem **h₂ur̥séh₂-*, which was later thematized as Indo-Iranian **ur̥ṣā- $\dot{i}á$ -ti*, featuring the accent on the thematic suffix as in regular deverbative or denominative presents. As for the syntax, one can compare the well-known construction of the dative sg. of verbal nouns in predicative function, such as Vedic RV *ná dābhāya* 'not to be deceived' (5.44.2c, 7.91.2a, 9.73.8a) or *ná (...) ādābhe* (RV 8.21.16d) with the same meaning. These negative phrases in nominal sentences are equivalent in the RV to the privative adjectives *ādabdha-*

²⁸Gotō 2013: 107. The type is reconstructed as *-ḡh₂- $\dot{i}e/o$ - by Jasanoff (2003: 123–124). The root *grabhⁱ* is currently traced back to **g^hrebh₂-* (LIV² 201), but the precise laryngeal in this present type does not matter in the present context.

²⁹Pace Gotō 2013: 131. In any case, one should renounce searching the origin of the factitive in the denominative of 'bull', through the metaphorical assimilation of the jet of sperm by the male to the flow of rain. This cannot be supported by the problematic etymological interpretation of PIE 'bull' itself, from a root that would mean 'to impregnate', *vel sim*.

³⁰Jamison 1983: 117–118.

³¹Scarlata 1999: 525.

(48x), *ádābh(i)ya-* (34x) ‘undeceivable’, *adábha-* (5.86.5b) ‘undeceptive’.³² Compare also the reading of the final dative of a root noun used as quasi-infinitive with factitive meaning in RV 1.89.5c *védasām ásat vṛdhé* ‘he will be for increasing/growing of possessions’ = ‘for letting grow the possessions’.³³

3.4 Consequences for Anatolian and PIE

The derivational scheme devised for explaining Ved. *vṛṣāyāti* applies in principle to any **-eh₂-* abstract dative sg. **X-eh₂-eḷ*. This was the precise basis of the so-called conversion. The predicative **[X-eh₂]-eḷ* in nominal sentences was reinterpreted as *X-eh₂-eḷ*, thus as 3sg. active present in **-e+i* of the **h₂e*-conjugation, with **X-eh₂-* reanalyzed as a denominative verbal stem. This happened after the 3sg. ending of the **h₂e*-conjugation was reanalyzed as an active ending, which yielded the Anatolian *hi*-conjugation. The further step to be made consists in transposing this process into PIE to generate the **-eh₂-*factitive present type. This can be supported by a concrete example found in Anatolian: **armaH-eḷ* ‘(in)to the moon’ ≈ ‘to become/make like the moon’ > Hitt. 3sg.act. *armaḥḥ-i* ‘makes N like the moon’ = ‘makes N pregnant’, see Section 3.1 above. This is consistent with the prevalent derivation of this factitive present type from abstract substantives, and by extension from other **-ah₂-* substantives, as shown by Sasseville (2021, see again Section 3.1). The form **armaH-eḷ* ‘(in)to the moon’ could in principle generate a labile verb, ‘to become/make (N) [round] like the moon’. But it was used transitively, in contrast to the essive/fientive suffixed denominative Hitt. *armā(i)-* ‘to be pregnant’ < **ah₂-ie/o-* ‘to be(come) [round] like the moon’. Hence, the former predicative was verbalized as Hitt. 3sg.act. *armaḥḥ-i* ‘makes N [round] like the moon’ > ‘makes N pregnant’. One should assume that the transitive reading prevailed for **-ah₂-*abstracts used predicatively in the dative singular with an overt nominal complement.

On the other hand, one cannot deny that the productivity of the type **neueh₂-* ‘to make new’ in Core IE as based on thematic adjectives had a primeval source in PIE, although its prevalence and extension to other types of adjectives is a phenomenon proper to Hittite in the Anatolian language family. One may surmise that this development was only embryonic in PIE. Nonetheless, it is likely that some **-eh₂-* abstracts could be derived from thematic adjectives. Therefore, the chain of derivation was extended with respect to the base, as in ex. (9).

³²On the relationship of infinitives and quasi-infinitives with verbal adjectives, see Renou 1937.

³³See Gippert 1978: 95, 96 and Hettrich & Stüber 2018: 23–25 for further instances.

- (9) adjective *néu-o- →
 abstract *néu-e-h₂- ‘newness’ →
 dative sg. *néu-eh₂-eĭ ‘for the purpose of newness, into newness’ →
 verb *néu-eh₂-e+i ‘(s)he makes X into newness’ > ‘(s)he makes X new’.

This derivational chain was then reduced to *néu-o- → *néueh₂-e+i, resegmented as *néu-e-h₂-e+i, in the deadjectival Hitt. *nēwahḫ*-type and its cognates in other IE languages. The restructuring of a complex chain by “jumping over” the intermediate link is a well-known phenomenon. The surface effect of the process was finally the apparent substitution of the nominal thematic vowel by the verbal suffix, thus contrasting with the *-o-ĭé/ó- or *-e-ĭé/ó- denominative presents.

This “de-datival” derivation of *h₂e-conjugation presents is to some extent corroborated by the derivation of Hitt. *hi*-verbs from adverbs, as described by Melchert (2009): *āppai*- ‘to go back, be finished’ besides *āppa* ‘after, (temporally) behind’, *parā*- ‘to appear, come/go forth’ (cf. HLuw. *ARHA para*- ‘go missing, be missing, lack’) from *parā* ‘forth, out’, *šanna*- ‘to conceal’ from **šanna*- ‘isolated, separated’ (< **sŋ(H)o*(-), cf. *šannapi* ‘in an isolated place’). An example of the latter is given in (10).³⁴

- (10) KBo 5.3. I 28
n-at-mu-kan *mān šannatti*
 CONJ-3SG.ACC.N-1SG.ACC/DAT-PTCL if conceal.2SG.PRS.ACT
n-at-mu *ŪL mematti*
 CONJ-3SG.ACC.N-1SG.ACC/DAT NEG tell.2SG.PRS.ACT
 ‘if you keep it secret from me, and do not tell me’

The secondary verbal inflection of local adverbs resides ultimately in the use of adverbs as predicate in nominal sentences.

The process underlying the formation of the *-eh₂-factitive is strikingly parallel to the derivation of the stative in *-eh₁- from a predicative adverb based on the instrumental sg. of a root noun, e.g. **h₁rud^h-éh₁* ‘with redness’ → stative **h₁rud^h-éh₁*- ‘to be red’, verbalized through addition of the present suffix *-ĭé/ó-³⁵ (see the introduction to this volume). In some languages, the stative itself coexisted with the productive formations of fientives in *-eh₁-s- and inchoatives in *-eh₁-sĕke/o-, or of periphrastic factitives, type Lat. *ārē-faciō* ‘to make dry’, beside

³⁴Further instances in Puhvel 2017: 111–113. The derivation proposed by Melchert (2009: 336–337) is more straightforward than through a verbal stem with a nasal infix (Oettinger 1979: 159–160; Kloekhorst 2008: 719) or with a nasal suffix (Puhvel 2017: 116).

³⁵See Jasanoff 2004: 145–149; also Merritt, this volume.

ārē-scō ‘to grow dry’. In addition, the stative present stem became the basis of nominal forms in **-eh₁-to-*, **-eh₁-ti-*, etc. The thematic renewal of the **-eh₂-* factitive type into **-eh₂-īe/o-* would be similar to the replacement of the **-eh₁-* stative type by **-eh₁-īe/o-*, e.g. in Lat. *rubeō*, *-ēre*, etc. Parallel to the formations issued from the stative, secondary adjectives in **-to-* and **-(e/o)nt-* based directly on the abstract probably also existed, which were later grammaticalized as participles of the newer **-ah₂-* factitives, to wit Lat. *(re)nouātus*, Hitt. *nēwahḫant-*.

4 Concluding remarks

In the first two sections of this paper, two alleged instances of denominative presents without an overt suffix have been falsified. This was made possible by resorting to more economic analogical processes. As an exemplary point, the theory of conversion of an adjective into a verb has been challenged in the case of **g^hih₃-uō-* ‘living’ and **g^hih₃-uē-ti* ‘lives’. This case suggests that the analogical process took place already in PIE. In the third section, the intriguing case of conversion apparently featured by the factitive **-eh₂-* present type has been treated in some detail and was resolved by resorting to a further analogical process in PIE. The crucial fact lies in the prevalent derivation of this factitive type from abstracts and individualizing derivatives, as still reflected by the Anatolian data. Furthermore, the later productive development of **-eh₂-* abstracts from thematic adnominal bases provides the missing link for deriving the **néu-eh₂-* type from thematic adjectives. The reconstructed process also explains without further ado the original **h₂e-* conjugation of the **-eh₂-* factitive type, which had been never accounted for. As an alternative approach, which would effectively be the admission of failure, one would be forced to assume that the verbal suffix **-eh₂-* of the present type and the nominal suffix **-eh₂-* had nothing in common and were simply homophonous. It does not seem commendable to multiply homophonous PIE morphemes only by projecting back the data on a purely formal basis without paying attention to the functions of the derivatives. This amounts to giving up on any perspective for a solution with the excuse of ignorance. With respect to the semantic side, the main issue concerned explaining the factitive meaning of the denominative **-eh₂-* present type, which follows from the syntactic reconstruction of predicatively used verbal abstracts with nominal complements.

Abbreviations

acc.	accusative	MH	Middle Hittite
act.	active	MP	Middle Persian
adj.	adjective	masc.	masculine
aor.	aorist	mid.	middle
Arm.	Armenian	Myc.	Mycenaean
AV	Śaṃhitās of the Atharvaveda	nom.	nominative
Av.	Avestan	nt.	neuter
CLuw.	Cuneiform Luwian	OAv.	Old Avestan
CToch.	Common Tocharian	OCS	Old Church Slavonic
dat.	dative	OH	Old Hittite
fem.	feminine	OHG	Old High German
Gaul.	Gaulish	OIr.	Old Irish
gen.	genitive	OP	Old Persian
Gk.	Greek	OPr.	Old Prussian
Go.	Gothic	Oss.	Ossetic
HLuw.	Hieroglyphic Luwian	PIE	Proto-Indo-European
Hitt.	Hittite	pl.	plural
IE	Indo-European	pres.	present
instr.	instrumental	ptcp.	participle
Lat.	Latin	RV	Śaṃhitā of the Ṛgveda
Latv.	Latvian	sg.	singular
Lesb.	Lesbian	Toch.	Tocharian
Lith.	Lithuanian	Ved.	Vedic
Lyc.	Lycian	VS	Vājasaneyi-Śaṃhitā
Lyd.	Lydian	YAv.	Young Avestan

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